

FTC Autonomous Lesson Focus Guide

Part 9 — Drive Forward Three Feet and Stop

Video discussion plus encoder calculation and measured floor trials

Use this guide to keep the lesson centered on three ideas: autonomous follows a prewritten sequence, encoders measure rotation rather than floor distance, and every moving-robot test requires the full POWERING ON call-and-clear sequence.

Before class

- [] Verify the actual motor, wheel, and drive-ratio constants.
- [] Tape the start, 36-inch target, boundaries, and stopping buffer.
- [] Mark one robot measurement reference point.
- [] Secure the robot on blocks for the first direction test.
- [] Print workbooks and display the POWERING ON poster.

Core vocabulary

autonomous | LinearOpMode | encoder | counts per revolution | wheel circumference | counts per inch | target position | RUN_TO_POSITION | isBusy | timeout | calibration

Success criteria

- Students calculate before testing.
- Students can name every independent stop path.
- Students never enter the lane until STOP and DISABLED are confirmed.
- Students measure from the same reference point.
- Students use repeated evidence before editing a constant.

Teacher pause questions

PAUSE	ASK	LISTEN FOR
0:09	What replaces the joystick in autonomous?	A prewritten sequence plus sensor feedback and stopping conditions.
0:18	What is our complete first mission?	Drive forward 36 inches, then command zero power.
0:29	Why must one drive motor be	Mirrored motors need opposite

PAUSE	ASK	LISTEN FOR
	reversed?	software directions to agree on physical forward.
0:39	What does an encoder measure—and what does it not measure?	Motor rotation; it does not directly know floor position or wheel slip.
0:49	How does wheel circumference connect counts to inches?	One wheel revolution travels one circumference, so counts per revolution can be divided by inches per revolution.
0:58	Why is the target about 1,630 rather than exactly 1,630?	The calculation produces 1,630.25 and the motor target must be a whole count.
1:09	What happens between POWERING ON and START?	Pause, step behind the boundary, recheck the lane, and receive the safety captain's clear.
1:29	What three commands begin the move?	Set target positions, select RUN_TO_POSITION, and apply low positive power.
1:39	Why do we need STOP and a timeout in addition to encoder targets?	They provide independent exits if the run is canceled or the mechanism cannot reach its target.
1:49	Does BRAKE make an enabled robot safe to approach?	No. It reduces coasting; STOP and confirmed DISABLED are still required.
2:00	Why measure several trials?	Real distance varies; repeated evidence supports calibration and reveals mechanical inconsistency.

Fast hands-on sequence

STEP	ACTION	STUDENT EVIDENCE
1	Audit	Confirm motors, wheels, reduction, names, and encoder cables while disabled.
2	Calculate	Convert 36 inches to approximately 1,630 encoder counts.
3	Trace	Put target, mode, power, monitor, timeout, and stop in order.
4	Clear	Run POWERING ON and clear the

STEP	ACTION	STUDENT EVIDENCE
		entire travel lane and buffer.
5	Test	Run one trial while everyone remains outside the boundary.
6	Disable	Press STOP and visually confirm DISABLED before entering.
7	Measure	Record actual distance, final counts, straightness, and surface.
8	Calibrate	Use repeated evidence, update one value, and confirm with another trial.

Common misconceptions

MISCONCEPTION	CORRECTION
A timeout controls distance	Encoder targets control the planned distance; timeout is a backup exit.
1,630 counts always equals 36 inches	It is a theoretical starting point for the stated hardware and ratio.
BRAKE locks the robot	BRAKE reduces coasting; it does not make approach safe.
Stopped motion means disabled	STOP and the Driver Station DISABLED indication are required.
One trial is enough to calibrate	Use several consistent trials and inspect mechanics when results scatter.